Distribution-Aware and Class-Adaptive Aggregation for Few-Shot Hyperspectral Image Classification

Ying Guo, Bin Fan, Yan Feng, *Member, IEEE*, Xiuping Jia, *Fellow, IEEE*, and Mingyi He, *Senior Member, IEEE*

***Abstract*— Recently, few-shot learning based on meta-learning has shown great potential in hyperspectral image classification (HSIC) due to its excellent adaptability to limited training samples. Despite achieving promising results, the existing methods ignore the interaction between the source domain (with abundant-labeled base-class samples) and the target domain (with few-labeled novel-class samples), as well as between the support set and the query set. This issue makes the resulting model usually biased toward the source domain and not robust to the sample variance of novel classes, posing a bottleneck to the improvement of HSIC performance. To overcome these limitations, we propose a flexible and effective distribution-aware and class-adaptive aggregation (DA-CAA) method for few-shot HSIC by transferring the class-level distribution information learned from the base classes to the novel classes. Specifically, we first employ a variational autoencoder (VAE), which is pretrained on abundant-labeled base-class samples, to encode the support set samples as class distributions. Subsequently, we sample class-level features from the learned distribution and adaptively aggregate them with sample-specific query features. This operation not only enhances cross-domain information interaction in a distribution-learning manner, but also ensures that the aggregated features across classes inherit both class-level and sample-specific information. Our proposed class-adaptive aggregation (CAA) encourages complementary fusion of features from all classes, which is beneficial for reducing class confusion. Experiments on four benchmark datasets demonstrate the effectiveness and flexibility of our approach.**

***Index Terms*— Feature aggregation, few-shot learning, hyperspectral image classification (HSIC), variational autoencoder (VAE).**

I. Introduction

HYPERSPECTRAL images (HSIs) are 3-D cubes obtained by using an imaging spectrometer to simultaneously acquire hundreds to thousands of bands within the same surface area. It holds significant importance in various fields such as mineral exploration [1], vegetation analysis [2], and target detection [3]. Particularly, HSI classification (HSIC), which advocated classifying objects by leveraging the spectral characteristics and spatial structural features of image pixels in different bands, was developed from multispectral image classification [4], [5]. Benefiting from the richer spectral and spatial information in HSIs, HSIC has gradually garnered widespread attention [6], [7], [8], [9], [10]. During the process of collecting HSI, factors such as environmental conditions and atmospheric interference can yield phenomena known as "different spectra for the same object" or "different objects with the same spectrum." As a result, obtaining discriminative and representative feature representations from HSI data has become a key element in achieving accurate HSIC.

In the early stages of research, HSIC methods primarily relied on manually designed feature classifiers in spectral dimensions, such as support vector machines (SVMs) [11], [12], multinomial logistic regression [13], [14], artificial immune system [15], [16], and dynamic subspace [17], [18]. Subsequently, spatial dimension information was gradually introduced to enhance the performance of classifiers, e.g., extended morphological profiles [19], [20], multiple kernels [21], [22], spatial–spectral kernels [23], [24], and filtering operations [25], [26] have been employed for this purpose. With the advancement of deep learning techniques, researchers recently have developed a series of effective methods for HSIC based on advanced neural network models [27]. For instance, He et al. [10] proposed using a deep stacked network to embed spectral and spatial features into a joint latent representation space, achieving reasonably good HSIC performance. Deep belief network [28] was used for learning the deep features of each land cover category. Moreover, recurrent neural networks [29] and generative adversarial networks [30] have also demonstrated great potential in HSIC. Domain adaptation-based HSIC methods are also developed [31], [32], [33], [34], [35], [36] for robust cross-domain classification. In particular, HSIC based on convolutional neural networks (CNNs) has been most widely studied. For example, HybridSN [37] and Fast3DCNN [38] utilized 2-D and 3-D convolution operations to efficiently explore spectral and spatial information for subsequent feature distinction. Multiscale 3-D CNNs [39], [40] have also been exploited to enable spatial and spectral feature extraction and fusion. These methods are capable of extracting efficient representation of spectral and spatial features in a data-dependent manner. Nevertheless, in most cases, the

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acquisition cost of HSI is high, and manual annotation relies on on-site inspections, which is time-consuming and labor-intensive. Therefore, the shortage of labeled HSI samples leads to a severe lack of prior knowledge, affecting the training of HSI feature extractors (FEs) and greatly limiting the practical deployment of the existing methods.

Instead of requiring a large number of labeled training samples, few-shot learning aims to learn a reliable model using limited labeled samples, thereby avoiding the high cost of data preparation and bringing hope for few-shot HSIC. Specifically, given a C-way K-shot classification task, the goal is to train a model that distinguishes C classes using only K labeled samples of each class [41], [42], where K is usually a small number (e.g., 1 or 5). The few-shot learning mainly consists of two training paradigms: data-driven and episode-based paradigms. The former takes advantage of prior knowledge from the source domain with abundant-labeled base-class samples, and then fine-tunes the model on the target domain with few-labeled novel-class samples, which facilitates the model's generalization ability to specified scenarios. For example, based on the idea of transfer learning, limited training samples of the current scene were adopted to learn top-level features [43], while low- and mid-level features are learned from other scenes. The latter can essentially be represented as a meta-learning task [36], [44], which takes the episode as the basic unit. Each episode is divided into a support set and a query set, and then the differences between samples are measured through meta-training and meta-testing, so that the model has the ability to learn independently of the episode, i.e., "learn to learn." Typically, the metric network model based on meta-learning can more intuitively express the intrinsic relationship between samples by learning the embedding representation of samples, and has become a mainstream learning paradigm. A multistage relation network with dual metric was constructed [45], which utilizes a sample recombination strategy to increase the variety of classification tasks. A few-shot HSIC method based on class-covariance metrics was presented [46] to learn a global representation for each episode through interactive training between the source and target domains, showing promising classification performance. The superior adaptability of meta-learning to limited training samples ensures that the HSIC model acquires meta-knowledge tailored for novel classes.

Despite the remarkable success of meta-learning in few-shot HSIC, the existing methods have not yet fully explored the information interactions between the source and target domains, as well as between the support and query sets, resulting in two obvious limitations. 1) The learned model is often biased toward the source domain. To complete knowledge transfer, the existing methods often parameterize a common HSI FE through interactive training between the source domain and the target domain [34], [46], [47], or by first training in the source domain and then fine-tuning in the target domain [48], [49], [50]. However, due to the significant disparity in number and appearance between base-class and novel-class samples, this imbalance can easily make the resulting model biased toward the source domain, leading to poor feature discriminability. Furthermore, since there are only K labeled samples for each novel class, the quality of the support set becomes crucial. As such, the performance is susceptible to the disturbance of the variance of support set samples, as shown in Fig. 1, resulting in poor model robustness. 2) Insufficient information interaction between the support set and the query set. To this end, one possible way is to aggregate query features with support features of the same class, i.e., class-specific aggregation (CSA) [34], [51]. This operation can embed more adequate class-specific information during training. However, the model may become confused during testing, as the matching support features cannot be known in advance. On the other path, class-random aggregation (CRA) [52] fuses query features with arbitrary support features, such that a class-agnostic representation is learned to reduce model confusion. However, randomly selected classes have limited complementary information, which is not conducive to embedding class-specific information.

To address the above problems, we propose a distribution-aware and class-adaptive aggregation (DA-CAA) method for few-shot HSIC. Our design philosophy is to efficiently deliver class-level distribution information learned from a large number of base-class samples to novel-class samples, aiming to narrow domain bias and improve the model robustness to the variance of few-shot samples. This is achieved by estimating class distribution with the widely used variational autoencoder (VAE) [53], which is beneficial to transforming sample-wise features into class-level features for robust feature aggregation. Specifically, we first employ VAE to model the feature distribution of the support set, i.e., training an effective VAE on a large number of base-class samples. Then, the pretrained VAE is extended to predict the feature distribution of few-labeled novel-class samples. Next, class-level features are sampled from the estimated feature distribution of the support set, which can enhance cross-domain information interaction capabilities. Since the estimated distribution is not biased toward specific samples, the features sampled from it are also robust to sample variances. Finally, the class-level features are fused with each sample-specific query feature through our proposed class-adaptive aggregation (CAA), thereby reliably transferring class-level distribution information to query features. Note that our CAA enables the query features to interact with the support features of all classes, thus integrating the common advantages of CSA and CRA, which can better model the interclass interaction and contribute to class confusion suppression. Experimental results on four benchmark datasets show that our approach has excellent HSIC performance compared to state-of-the-art methods. Moreover, when adapting our pipeline to a variety of HSI feature extraction baselines, consistent HSIC improvements are achieved, demonstrating the effectiveness and flexibility of our approach.

The main contributions of this article can be summarized as follows.

1) We devise a tractable distribution-aware feature embedding (DAFE) for few-shot HSIC, which transforms sample-wise support features into class-level features, facilitating the transduction of class distribution information in a cross-domain interaction manner.

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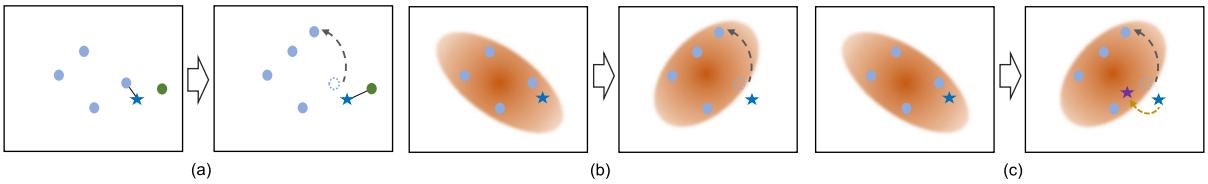


Fig. 1. Comparisons of different feature encoding methods. Blue dots denote support features, green dots denote support features of another class, and blue pentagrams are query features. When the characteristics of one of the support features change significantly (see gray dashed arrows), indicating an obvious alter in variances of few-shot samples, both (a) point estimation methods [41], [54] and (b) class-covariance estimation methods [46], [47] (obtained by direct calculation based on current samples) may lead to class confusion, indicating that the resulting model is not robust to the variance of support set samples. In contrast, (c) our method aims to model support set samples as class distributions and inject them into query features through *cross-domain and cross-class adaptive interactions* to obtain *enhanced query features* (purple pentagram) with *better distinguishability*, thus enjoying higher robustness to varying variances.

2) We propose a simple yet effective CAA module, which enhances the reliability of model predictions through sufficient cross-class feature interactions between the support set and the query set.
3) Extensive experiments on four public HSIC datasets show that our approach surpasses the previous methods effectively, setting a new state-of-the-art.

The rest of this article is organized as follows. Section II presents the prior knowledge of few-shot HSIC. Section III describes the proposed method in detail. Experimental details and results on four public HSI datasets are presented in Section IV. At last, a conclusion is drawn in Section V.

II. Preliminaries

To better understand the method proposed in this article, several important concepts are first explained. The dataset for few-shot learning usually contains the source domain datasets D_s with C_s classes and the target domain datasets D_t with C_t classes, where $D_s \cap D_t = \phi$. The source domain advocates providing prior knowledge for the target domain because, in principle, D_s has significantly more labeled samples than D_t. The training strategy usually adopts the episode-based approach, i.e., the meta-learning task [36], [44], [55], in which each episode is divided into a support set $\mathcal{S}$ and a query set $\mathcal{Q}$. As such, an episode corresponds to a C-way K-shot task, where C represents the number of classes randomly selected from the training data and K refers to the number of labeled samples per class. Each class of the support set will contain K labeled samples, denoted as $\mathcal{S} = \{(x_i, y_i)\}_{i=1}^{C\times K}$. The query set draws N samples from each of the same C classes as the support set, denoted as $\mathcal{Q} = \{(x_j, y_j)\}_{j=1}^{N\times K}$. In this way, we extract samples from D_s based on C classes to obtain the base-class samples for model training. Note that the base-class samples are all abundantly labeled. During meta-training, the model learns from abundant-labeled base-class samples in D_s and then generalizes the learned prior knowledge to the meta-task of few-labeled novel-class samples in D_t. Additionally, due to the limited labeled samples in the novel classes, in order to form a sufficient training set for model fine-tuning, following [46], [47], we augment the target domain data with randomly added Gaussian noise, as shown in Fig. 2. As a result, a meta-learning task is constructed in the target domain in a way similar to that of the source domain.

III. Method

In this section, we first provide an overview of the overall framework of our proposed DA-CAA method in Section III-A. Then, we sequentially introduce three key modules, i.e., HSI FE in Section III-B, DAFE in Section III-C, and CAA in Section III-D, followed by an explanation of the used metric calculation scheme in Section III-E.

A. Overall Framework

Considering thje insufficient class information interaction in the existing methods, which often leads to a bias toward base classes and is not robust to the variance of novel-class samples, we propose an effective and flexible DA-CAA in this article. The overall network framework is illustrated in Fig. 2. Its motivation is to efficiently transfer the class-level distribution information learned from a large number of base-class samples to novel-class samples by modeling the distribution of each class in the support set, thereby alleviating these limitations. First, we employ an HSI FE to map data from both the source and target domains into a unified representation space. Then, we train a VAE on a large number of base-class samples to encode the support set samples into class distributions and sample class-level features from them. This constitutes our DAFE module, where the estimated class distributions are unbiased toward specific samples, favoring the generation of more robust predictions. Subsequently, the class-level features interact with the sample-specific query features through a CAA module, thus reliably integrating class-level distribution information to query features. Finally, the enhanced query features, which integrate both crucial class-level and sample-specific information, are fed into a tractable metric calculator to perform feature discrimination. In addition, we further fine-tune the model parameters in the target domain, where the DAFE module is frozen to retain adequate prior knowledge. In the inference phase, for unlabeled novel-class samples, we directly perform category prediction using the trained model parameters, where class-level features are embedded to reduce class confusion and upgrade the model's robustness.

B. HSI Feature Extractor

Although HSI brings richer and more direct spectral information, the spectral bands within HSI often exhibit redundancy because they may convey similar information. Therefore, harnessing an effective HSI FE $\mathcal{F}$ becomes crucial for extracting discriminative and representative features from HSI data, which will be beneficial for improving the reliability of class distribution. Note that our DA-CAA is compatible with arbitrary HSI FEs, which ensures the flexibility of our method, as shown in Section IV-D. In this article, we adopt our

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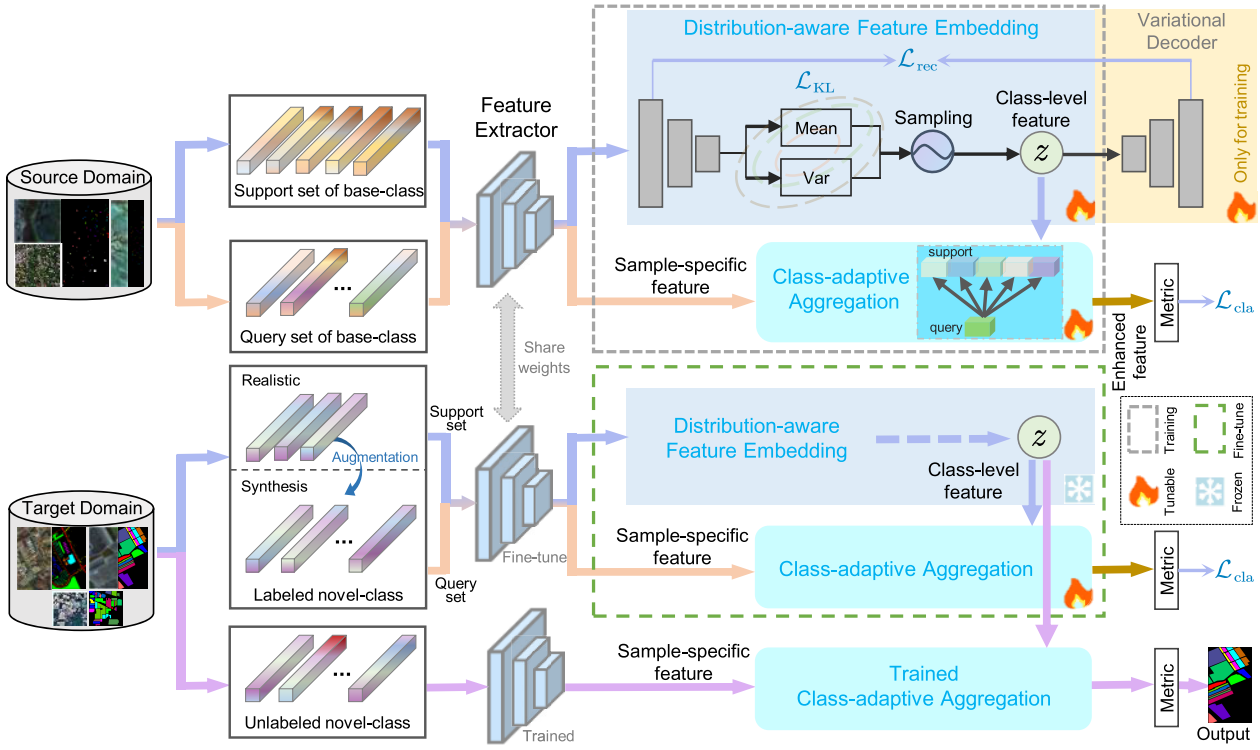


Fig. 2. Overall framework of our proposed DA-CAA method. In the source domain, we first extract sample-wise features from both the support set and the query set using a weight-shared HSI FE. Then, we train a VAE on abundant-labeled base-class samples to encode the support set into class distributions. Note that the variational decoder is only used during training. Afterward, we sample class-level features from the learned distributions. Finally, we adaptively aggregate them with sample-specific query features to reconstruct enhanced query features for HSIC. In the target domain, we perform parameter fine-tuning, where we freeze the DAFE module to retain more sufficient prior knowledge. Similarly, for unlabeled novel-class samples, we adopt the trained parameters and embed class-level features to produce the desired outputs.

recently proposed Grid-TF [47] as the HSI FE. Notably, Grid-TF utilizes a powerful transformer [56] to compute nonlocal self-similarity along the spectral dimension and integrates this capability into a novel grid-like network architecture, which has been proven to excel in capturing fine features within HSI. The HSI FE is weight-shared for the support set and the query set.

Since different HSI data typically have different spectral band lengths, in order to reduce spectral redundancy and guarantee uniform processing, we perform a band selection operation on the original HSI before feeding it to the HSI FE. Specifically, we first uniformly select 100 spectral bands by employing the OCBS method [57]. Then, similar to [46] and [47], a 9×9 image patch $I_i \in \mathbb{R}^{9\times9\times100}$ centered on pixel (x_i, y_i) of each sample is input into the HSI FE. According to the meta-task settings, the data in the source and target domains are entered into the network training with a support set and a query set, respectively. The weight-shared HSI FE abstracts the query set and the support set into L-dimension feature representations, respectively, yielding the support features as $f^{\mathcal{S}} \in \mathbb{R}^{CK\times L}$ and the query features as $f^{\mathcal{Q}} \in \mathbb{R}^{NK\times L}$, i.e.,

$$f^{\mathcal{S}} = \{\mathcal{F}_{\mathcal{S}}(I_i)\}_{i=1}^{C\times K}, \quad f^{\mathcal{Q}} = \left\{\mathcal{F}_{\mathcal{Q}}\left(I_j\right)\right\}_{j=1}^{N\times K}. \tag{1}$$

C. Distribution-Aware Feature Embedding

Previous few-shot works usually encode few samples into discrete points [41], [54] or class-covariances [46], [47], as shown in Fig. 1(a) and (b), which poses challenges in effectively representing the class distribution. The metric of models is unreliability when the number of support sets is insufficient (i.e., few-shot) and their quality is uneven (i.e., varying variances). Furthermore, the existing methods heavily rely on reusing features of base classes within the feature embedding space learned from the source domain, which exacerbates domain bias during model training due to the imbalance in the number of labeled samples of base classes and novel classes.

To reduce the bias of the model toward specific samples, we develop a tractable DAFE module to transform sample-wise support features into class-level features for cross-domain and cross-class feature aggregation. First, we train a widely used VAE [53] on abundance-labeled base-class samples to encode the support features $f_{i^*}^{\mathcal{S}} \in \mathbb{R}^{1\times L}$ of each sample $i^* \in [1, CK]$ into a class distribution $\mathcal{N}$. Afterward, we sample the corresponding latent variables z_{i^*} from $\mathcal{N}$, followed by an averaging operation for each class to produce $\mathcal{Z} = \{z_i\}_{i=1}^{C}$ as the class-level features, which can be exploited for robust feature aggregation in a distribution-learning manner. Since $\mathcal{Z}$ contains more general class features and is not biased toward specific samples, this processing ensures better robustness to the variance of the support set samples [see Fig. 1(c)]. That is, extracting class-level distribution information by our DAFE is beneficial, as demonstrated in Section IV-E4. Notably, we use the variational encoder of VAE to implement our DAFE, while its variational decoder is only

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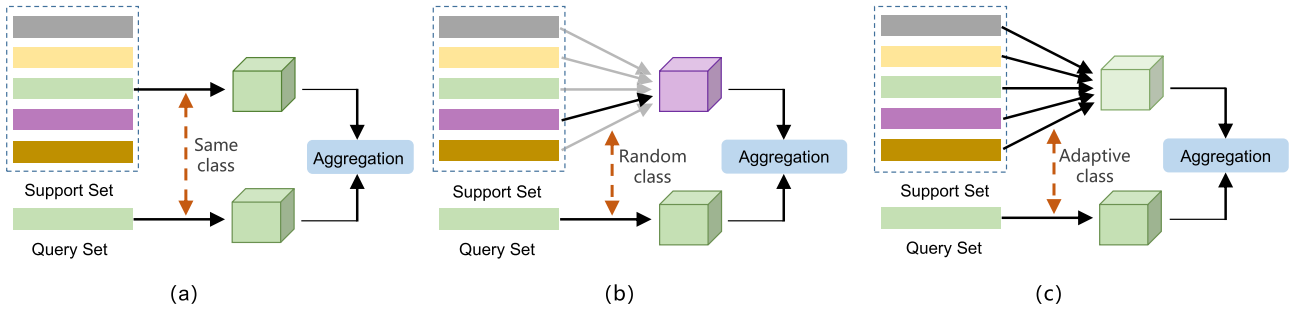


Fig. 3. Comparison of three feature aggregation paradigms. (a) CSA aims to aggregate query features with support features of the same class. (b) CRA fuses query features with arbitrary support features. (c) Our CAA encourages query features to interact with support features from all classes, which integrates the benefits of CSA and CRA to mitigate class confusion.

used to learn a reliable model during the training phase. Then, we will elaborate on these steps in detail, and the subscript i is dropped for ease of presentation.

Given the support feature $f^{\mathcal{S}}$, we optimize the model parameters in a similar manner to VAE, but our objective is to sample the latent variable z instead of reconstructing the support feature $f'^{\mathcal{S}}$. z can map the support feature $f^{\mathcal{S}}$ into a continuous space encapsulating distribution information. According to the principles of VAE, we assume that z is generated by the prior distribution $p(z)$, and $f^{\mathcal{S}}$ is reconstructed from a conditional distribution $p(f^{\mathcal{S}}|z)$ in the variational decoder. The process of modeling $f^{\mathcal{S}}$ is hidden, that is, z is unknown, and thus variational inference can be used to model the posterior distribution $p(z|f^{\mathcal{S}})$. The true posterior distribution $p(z|f^{\mathcal{S}})$ can be approximated with another distribution $q(z|f^{\mathcal{S}})$. The Kullback–Leibler (KL) divergence between the true posterior distribution and its approximation can be expressed as follows:

$$\mathrm{KL}\big[q(z|f^{\mathcal{S}})\|p(z|f^{\mathcal{S}})\big] = \int_z q(z|f^{\mathcal{S}}) \log \frac{q(z|f^{\mathcal{S}})}{p(z|f^{\mathcal{S}})}. \tag{2}$$

Because the KL divergence is always greater than or equal to zero, maximizing the marginal likelihood $p(f^{\mathcal{S}})$ is equivalent to maximizing the evidence lower bound (ELBO), i.e.,

$$\mathrm{ELBO} = \mathbb{E}_{q(z|f^{\mathcal{S}})}\big[\log\big(p(f^{\mathcal{S}}|z)\big)\big] - \mathrm{KL}\big(q(z|f^{\mathcal{S}})\|p(z)\big). \tag{3}$$

Similar to [52] and [58], we set the prior distribution of z to be a centered isotropic multivariate Gaussian, i.e., $p(z) = \mathcal{N}(0, I)$. Furthermore, we assume the posterior distribution $q(z|f^{\mathcal{S}})$ to be a multivariate Gaussian with diagonal covariance, defined as follows:

$$q(z|f^{\mathcal{S}}) = \mathcal{N}(\mu, \sigma) \tag{4}$$

where μ and σ can be estimated by the variational encoder. As a rusult, the variational feature (i.e., the class-level feature) z can be sampled by using the reparameterization trick [53], i.e.,

$$z = \mu + \sigma \cdot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I). \tag{5}$$

To predict the maximum likelihood $p(f^{\mathcal{S}}|z)$, we employ a variational decoder to reconstruct the original support feature $f'^{\mathcal{S}}$ from z and minimize the $\mathcal{L}_2$ distance between the original $f^{\mathcal{S}}$ and the reconstructed $f'^{\mathcal{S}}$. Therefore, a reconstruction loss $\mathcal{L}_{\mathrm{rec}}$ can be derived to represent the first term of (3), i.e.,

$$\mathcal{L}_{\mathrm{rec}} = \big\|f^{\mathcal{S}} - f'^{\mathcal{S}}\big\|. \tag{6}$$

Moreover, we directly minimize the KL divergence to model the second term of (3), i.e.,

$$\mathcal{L}_{\mathrm{KL}} = \mathrm{KL}(q(z|\mathcal{S})\|p(z)) \tag{7}$$

which encourages z to follow a Gaussian distribution.

In summary, the DAFE loss $\mathcal{L}_{\mathrm{DAFE}}$ can be expressed as the linear weighting of the reconstruction loss $\mathcal{L}_{\mathrm{rec}}$ and the KL divergence loss $\mathcal{L}_{\mathrm{KL}}$ so that the model can successfully learn the underlying distribution of the support set. Formally

$$\mathcal{L}_{\mathrm{DAFE}} = \mathcal{L}_{\mathrm{rec}} + \lambda_{\mathrm{KL}}\mathcal{L}_{\mathrm{KL}} \tag{8}$$

where λ_{KL} is a regularization parameter to balance the reconstruction quality and underlying distribution. We set λ_{KL} as a default value of 2.5×10^{-4}, similar to [52] and [59], to ensure that DAFE is capable of extracting meaningful information from the input. For the HSIC task, we further use the cross-entropy loss as our classification loss $\mathcal{L}_{\mathrm{cls}}$. In conclusion, the total loss function $\mathcal{L}_{\mathrm{total}}$ can be formulated as follows:

$$\mathcal{L}_{\mathrm{total}} = \lambda_{\mathrm{cls}}\mathcal{L}_{\mathrm{cls}} + \mathcal{L}_{\mathrm{DAFE}} \tag{9}$$

where the hyperparameter λ_{cls} is set to 5 to balance distribution information extraction and object classification.

D. Class-Adaptive Aggregation

Feature aggregation has been considered an effective method for feature interaction in meta-learning [52], [60], [61]. There are two existing feature aggregation methods, one is the CSA [34], [51], and the other is the CRA [52], as depicted in Fig. 3. Essentially, CSA attempts to aggregate query features with support features of the same class, which can introduce richer class-specific information during the training phase. Nevertheless, in the inference phase, the corresponding support features become unknown, which will lead to model confusion. CRA aims at integrating query features with support features from random classes, which can reduce model confusion to some extent. However, randomly selected classes possess limited supplementary information, which hinders the aggregation of class-specific information and makes network training difficult. In contrast, to reliably integrate class-level distribution

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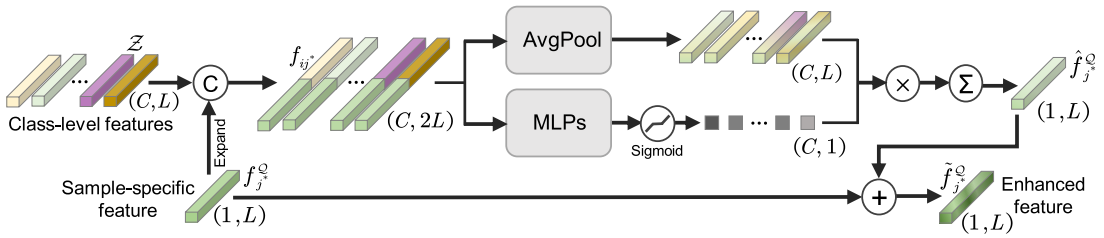


Fig. 4. Network details of the proposed CAA module. Given a sample-specific feature in the query set, we first concatenate it with the class-level feature. Then, we perform intraclass interaction through average pooling, while utilizing multiple MLPs combined with the sigmoid function to estimate aggregation weights. Subsequently, they perform element-wise multiplication and interclass addition to estimate complementary information corresponding to this query set sample. Finally, enhanced query features are generated by feature addition. In this way, the query features interact sufficiently with features from each class, facilitating better feature discrimination in subsequent steps.

information to query features, we devise a simple yet effective CAA module, as shown in Fig. 4, which encourages query features to interact with support features of all classes. This design can enhance the robustness of feature discrimination through sufficient cross-class feature interactions between the support set and the query set. Notably, it is easy to find that CSA and CRA are special cases of our CAA. Therefore, our CAA can simultaneously benefit from the common advantages of CSA and CRA, which not only facilitates the adaptive introduction of more comprehensive class-specific information but also faithfully alleviates model confusion and boosts model robustness, as exemplified in Section IV-E1.

The goal of our CAA is to adaptively aggregate the sample-specific query feature $f_{j^*}^{\mathcal{Q}} \in \mathbb{R}^{1\times L}, 1 \le j^* \le NK$ and the class-level features $\mathcal{Z} = \{z_i\}_{i=1}^{C} \in \mathbb{R}^{C\times L}$ to form an enhanced query feature $\tilde{f}_{j^*}^{\mathcal{Q}}$, i.e.,

$$\tilde{f}_{j^*}^{\mathcal{Q}} = \mathcal{A}\left(f_{j^*}^{\mathcal{Q}}, \mathcal{Z}\right), \quad j^* \in [1, NK] \tag{10}$$

where $\mathcal{A}$ represents our CAA module. Specifically, we first expand the dimensions of $f_{j^*}^{\mathcal{Q}}$ to $C \times L$, and then concatenate it with each class-level feature $z_i \in \mathbb{R}^{C\times L}$, $1 \le i \le C$ in the channel dimension in turn, yielding a cascaded feature representation f_{ij^*} with a dimension of $C \times 2L$. Then, average pooling is employed to achieve intraclass interactions. Meanwhile, f_{ij^*} is probabilized by using multilayer perceptrons (MLPs) and the sigmoid function to generate appropriate weights for feature aggregation. On this basis, they are element-wise multiplied, followed by interclass addition to produce the complementary feature $\hat{f}_{j^*}^{\mathcal{Q}}$ corresponding to the query feature $f_{j^*}^{\mathcal{Q}}$. Formally

$$\hat{f}_{j^*}^{\mathcal{Q}} = \sum_{i=1}^{C}\left\{\left(\text{AvgPool}\left(f_{ij^*}\right) \odot \text{Sig}\left(\text{MLPs}\left(f_{ij^*}\right)\right)\right)\right\}. \tag{11}$$

Finally, we obtain the enhanced query feature $\tilde{f}_{j^*}^{\mathcal{Q}}$ through residual connection, i.e.,

$$\tilde{f}_{j^*}^{\mathcal{Q}} = f_{j^*}^{\mathcal{Q}} + \hat{f}_{j^*}^{\mathcal{Q}}. \tag{12}$$

This process is similarly performed for all query features, i.e., $j^* \in [1, NK]$, resulting in enhanced query features $\tilde{f}^{\mathcal{Q}}$. In this manner, we achieve sufficient cross-class interaction between the query set and the support set, which helps propagate class-level distribution information to query features to model class relations and improve feature discriminability.

E. Metric Calculation

We model the support set as class distributions through our DAFE in Section III-C, and transmit the class-level distribution information to the new classes of the query set through our CAA in Section III-D. These cross-domain and cross-class information interactions ensure that the embedded features contain distribution information, which naturally fits well with the Mahalanobis distance (MD). This is because the MD considers the correlation between classes to determine the decision boundaries, while our enhanced query features contain sufficient distribution information, which are helpful in estimating a more accurate covariance matrix Σ. Therefore, instead of using the standard Euclidean distance (ED), we calculate the MD to formulate the objective metric function for each episode as follows:

$$\hat{\theta} = \arg\min_{\theta} \sum_{i=1}^{C}\sum_{j=1}^{NK}\left\{\left(\tilde{f}_j^{\mathcal{Q}} - \mu_i\right)\Sigma^{-1}\left(\tilde{f}_j^{\mathcal{Q}} - \mu_i\right)^{\mathrm{T}}\right\} \tag{13}$$

where θ denotes the learnable network parameter of our proposed DA-CAA classifier, and $\mu_i \in \mathbb{R}^{1\times L}$ is the mean of the support features $f_i^{\mathcal{S}} \in \mathbb{R}^{K\times L}$ of each class $i \in [1, C]$. The effectiveness of this metric scheme is confirmed in the experiments of Section IV-E5.

IV. Experiments

A. Datasets

The datasets for few-shot HSIC can be divided into source domain and target domain. In our experiments, the source domain involves three diverse datasets, namely Chikusei[1] (CKS), Kennedy Space Center[2] (KSC), and Botswana[2] (BW). Accordingly, the target domain contains University of Pavia[2] (UP), Salinas[2] (SA), Indian Pines[2] (IP), and Houston 2018[3] (HT18) datasets. The category information is shown in Tables I and II, respectively. We perform experimental evaluation and ablation analysis on target domain datasets.

1) Source Domain Datasets: The details of the source domain datasets are as follows:

1) The CKS dataset was collected over agricultural and urban areas in Chikusei, Japan, by the Headwall Hyperspec-VNIR-C sensor with a spatial resolution of

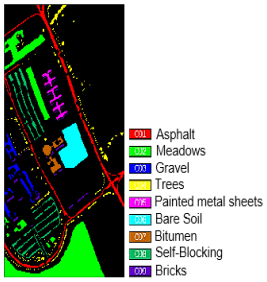
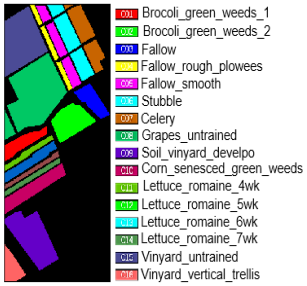
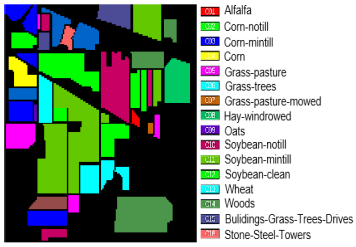
[1]https://naotoyokoya.com/Download.html
[2]https://rslab.ut.ac.ir/data
[3]https://hyperspectral.ee.uh.edu/?page_id = 1075

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TABLE I
CLASS TYPES OF SOURCE DOMAIN DATASETS

Datasets	Sample Num.	Class Types
CKS	77592	1) Water, 2) Bare Soil(school), 3) Bare Soil (park), 4) Bare Soil (farmland), 5) Natural Plants, 6) Weeds Infarmland, 7) Forest, 8) Grass, 9) Rice field (grown), 10) Rice field (first stage), 11) Row crops, 12) Plastic House, 13) Manmade (non-dark), 14) Manmade (dark), 15) Manmade (blue), 16) Manmade (red), 17) Manmade Grass, 18) Asphalt, 19) Paved Ground
KSC	5211	1) Scrub, 2) Willow Swamp, 3) CPhammock, 4) Slashpine, 5) Oak/Broadleaf, 6) Hardwood, 7) Swamp, 8) Graminoid Marsh, 9) Spartina Marsh, 10) Cattail Marsh, 11) Saltmarsh, 12) Mudflats, 13) Water
BW	3248	1) Water, 2) Hippo Gras, 3) Floodplain Grass1, 4) Floodplain Grass2, 5) Reeds1, 6) Riparian, 7) FireScar2, 8) Island Interior, 9) Acacia Woodlands, 10) Acacia Shrublands, 11) Acacia Grasslands, 12) Short Mopane, 13) Mixed Mopane, 14) Exposed Soils

TABLE II
GT AND THE CORRESPONDING CLASS TYPES OF TARGET DOMAIN DATASETS

Pavia University (UP)	Salinas (SA)	Indian Pines (IP)	Houston 2018 (HT18)
			

2.5 m. The data contains 128 bands in the wavelength range of 363∼1018 nm. The image size is 2517 × 2335 pixels, containing 77 592 samples, covering 19 types of ground objects.

2) The KSC dataset was collected at the Kennedy Space Center in Florida through the NASA AVIRIS sensor. The original data contains 224 bands in the wavelength range of 400∼2500 nm. After removing water absorption and low signal-to-noise ratio bands, 176 bands are reserved. The image size is 614 × 512 pixels with a spatial resolution of 18 m, containing 5211 samples, covering 13 types of vegetation objects.

3) The BW dataset was collected at the Kennedy Space Center in Florida through the EO-1 sensor with a spatial resolution of 30 m. The original data has 242 bands covering the wavelength range of 400∼2500 nm in 10 nm windows. After removing uncalibrated bands and noise bands covering water absorption features, the remaining 145 bands are used for analysis. The image size is 1476 × 256 pixels, containing 3248 samples, covering 14 types of land cover objects in seasonal swamps, occasional swamps, and drier woodlands.

2) Target Domain Datasets: The details of the target domain datasets are as follows:

1) The UP dataset was collected at the University of Pavia in northern Italy through ROSIS sensors, with a spatial resolution of 1.3 m. About 42 776 samples with 103 bands are incorporated. The image size is 610 × 610 pixels, and the landscape includes buildings and streets.

2) The SA was collected in Salinas, California, using the AVIRIS sensor with a spatial resolution of 3.7 m. The original data contains 224 bands in the wavelength range of 400∼2500 nm. After removing the uncalibrated bands with water absorption features, the remaining 204 bands are used for analysis. The image size is 512 × 217 pixels, containing 54 129 samples, covering 16 types of ground objects. The landscape mainly includes vegetables, bare soil, and botanical gardens.

3) The IP dataset was collected in Indiana using the AVIRIS sensor with a spatial resolution of 20 m. The original data possesses 224 bands in the wavelength range of 400∼2500 nm. After removing the uncalibrated bands and noise bands covering the water absorption characteristics, the remaining 200 bands are retained for analysis. The image size is 145 × 145 pixels, containing 10 249 samples, covering 16 types of ground objects. The landscape comprises two-thirds of agriculture and one-third of forests or other natural perennial plants.

4) The HT18 dataset was acquired over the University of Houston campus and the neighboring urban area from the IEEE GRSS DFC in 2018. The image size is 601 × 2384 with a spatial resolution of 1 m. It comprises 48 spectral bands in the wavelength range of 380∼1050 nm with 504 856 samples. Among them, 20 categories are contained, ranging from various vegetation types to urban infrastructure, which is significantly different from the landscape in the source domain. Note that we merely employ the HT18 dataset to evaluate the classification accuracy of our method in Section IV-C.

B. Experimental Details

We integrate the source domain datasets so that the network can obtain reliable distribution information. In particular, from

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TABLE III
CLASSIFICATION RESULTS (%) OF VARIOUS METHODS WITH FIVE LABELED SAMPLES PER CLASS ON THE UP DATASET

No.	SVM [11]	HybridSN [37]	Fast3DCNN [38]	DCFSL [36]	Gia-CFSL [48]	DM-MRN [45]	Grid-TF [47]	**Ours**
1	65.97±13.37	56.95±15.07	68.17±9.85	76.07±10.09	79.99±13.67	79.36±8.51	79.21±10.46	**81.98±7.19**
2	57.08±30.61	78.59±8.76	81.14±9.45	83.71±7.59	82.73±8.62	75.91±9.81	82.80±8.70	**86.22±6.93**
3	56.12±15.14	60.68±13.54	56.72±11.23	57.44±12.94	64.27±10.43	**79.37±8.41**	73.08±11.42	72.73±10.66
4	86.30±7.93	89.92±3.80	91.63±3.83	90.13±5.52	92.51±2.94	88.84±10.06	93.28±2.59	**93.31±3.29**
5	98.28±2.84	99.43±1.02	98.25±3.17	98.01±3.07	99.45±0.60	**99.78±0.35**	99.64±0.51	99.77±0.31
6	45.40±25.02	59.08±11.97	59.38±11.32	68.40±12.88	76.52±12.95	**85.62±16.01**	73.22±10.96	70.04±11.72
7	53.46±24.69	81.06±9.51	82.94±6.94	75.90±9.95	83.06±6.88	**98.92±0.61**	80.29±9.04	88.32±5.94
8	30.49±17.54	57.75±16.89	64.98±18.54	67.61±13.29	77.13±13.30	75.75±10.39	75.59±11.80	**84.00±11.56**
9	94.79±5.25	95.56±6.29	96.18±3.37	98.18±1.86	98.76±1.85	**99.38±0.51**	96.42±3.44	96.49±4.25
AA	65.33±2.36	75.45±1.71	77.71±1.51	79.49±3.06	83.82±2.09	**87.17±2.45**	83.73±1.48	85.87±2.50
OA	58.85±10.15	72.18±3.92	75.66±3.78	79.03±2.99	81.77±4.31	77.30±5.91	81.98±3.38	**84.03±2.67**
Kappa	49.54±8.89	64.33±4.36	68.59±4.19	72.83±3.54	76.57±4.98	71.82±6.77	75.55±3.92	**79.24±3.20**

the source domain datasets containing 46 classes, 40 classes including a sufficient number of labeled samples are filtered out to form the base-class samples, ensuring that enough prior knowledge can be learned during the meta-training phase. In the C-way K-shot task, we select 180 support set samples and 19 query set samples for each class, which is similar to [46]. Notably, since the target domain datasets only contain limited labeled samples, we augment the number of support set samples from K to 180 by randomly adding Gaussian noise for parameter tuning. Unless otherwise stated, we set K to 5 (i.e., five-shot). In this setting, five labeled samples are randomly selected from each category in the target dataset to construct the support set, while the remaining samples are used as the query set for model testing. Furthermore, C is determined based on the number of sample classes C_t in the target domain datasets. Specifically, when classifying UP, SA, IP, and HT18 datasets, C is set to 9, 16, 16, and 20, respectively. This setup can facilitate cross-domain learning in a shareable distribution space, similar to the previous works, such as [36], [46], and [48].

To optimize the parameters of the model, the Adam optimizer [62] with a learning rate of 0.001 is adopted. The feature dimension L of the class-level features $\mathcal{Z}$ is set to 128. We train our method on the source domain with abundant-labeled base-class samples for 8000 iterations to encode the support features into reliable class distributions. Also, 2000 iterations of parameter tuning are performed on the target domain with few-labeled novel-class samples. Note that our DAFE module is frozen during fine-tuning to preserve more holistic distribution information. At the same time, in order to avoid experimental accidents, we conduct ten independent experiments to verify the stability and reliability of the network. Throughout the experiments, a single NVIDIA GeForce RTX 3090 GPU is harnessed for model training and inference.

For evaluation metrics, we adopt class-specified classification accuracy (CA), average classification accuracy (AA), overall classification accuracy (OA), and kappa coefficient (Kappa), which have been widely utilized in the HSIC task, to comprehensively evaluate the performance of the models. Higher metric values indicate better HSIC performance.

C. Comparison With Existing Methods

To validate the performance of the proposed method, we compare the classification results with state-of-the-art methods, including traditional, and learning-based methods. As a traditional method, SVM [11] uses kernel functions to map HSI data into a high-dimensional space to make the features easier to distinguish. The contrasting few-shot HSIC methods based on deep learning consist of HybridSN [37], Fast3DCNN [38], DCFSL [36], Gia-CFSL [48], DM-MRN [45], and Grid-TF [47]. Concretely, HybridSN and Fast3DCNN design effective spatial–spectral FEs based on convolution operations. In particular, Fast3DCNN is a deep network based on 3-D convolution, and HybridSN is constructed from 2-D and 3-D convolution. Furthermore, DCFSL and Gia-CFSL aim to solve the cross-domain problem between the source and target domains through domain alignment to improve the accuracy of HSIC. DM-MRN attempts to represent the class distribution of few-labeled samples by designing image-to-class and image-to-image blocks. Tables III–V report the average accuracy and standard deviation on the UP, SA, and IP datasets, respectively.

Table III presents the classification performance of various methods on the UP dataset under the five-shot condition. It can be observed that the performance of deep learning algorithms far exceeds the traditional SVM method. Overall, our approach achieves the best results on OA and Kappa and the second-best performance on AA. DM-MRN [45] focuses more on extracting and representing the features of each class separately, which makes it easy to improve the classification accuracy of each class, thereby contributing to better AA results. However, DM-MRN has inferior OA and Kappa metrics, which is due to the fact that it cannot pay attention to the overall data distribution well, indicating an underlying bias in model classification when there is a significant class imbalance in the target domain dataset. Similarly, this phenomenon can also be observed in Tables IV and VI. Compared with cross-domain few-shot HSIC methods DCFSL and Gia-CFSL, our approach surpasses them in terms of AA, OA, and Kappa by 8.03%, 6.33%, 8.80%, and 2.45%, 2.76%, 3.49%, respectively. It is also worth mentioning that our approach also has relatively smaller classification prediction variances. These results demonstrate that the class distributions estimated by ours are able to narrow domain bias and refine the model robustness to the variance of few-shot samples. In addition to verifying the effectiveness of our method on the UP dataset containing urban buildings and roads, we next test it on SA and IP datasets with agricultural information and on the HT18 dataset with vegetation and urban infrastructure. In particular, achieving the classification of IP and HT18 datasets is challenging.

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TABLE IV
CLASSIFICATION RESULTS (%) OF VARIOUS METHODS WITH FIVE LABELED SAMPLES PER CLASS ON THE SA DATASET

No.	SVM [11]	HybridSN [37]	Fast3DCNN [38]	DCFSL [36]	Gia-CFSL [48]	DM-MRN [45]	Grid-TF [47]	**Ours**
1	71.16±31.20	99.05±1.30	99.48±0.69	99.01±1.02	99.62±0.33	**99.88±0.24**	99.62±0.50	98.86±1.31
2	81.23±24.57	97.60±1.69	99.76±0.63	99.16±1.08	99.18±1.23	**99.86±0.17**	99.85±0.27	99.59±0.44
3	54.18±11.39	91.91±5.93	**94.02±6.40**	91.81±8.20	93.60±9.05	93.38±14.10	87.43±12.88	89.67±9.80
4	97.18±1.22	99.36±0.49	99.60±0.27	**99.67±0.38**	99.11±1.40	99.12±1.71	98.39±1.44	99.53±0.55
5	86.09±10.62	89.85±3.91	92.34±3.26	91.10±3.86	92.71±3.67	95.20±4.95	92.53±6.84	**94.38±4.34**
6	97.69±1.75	97.94±1.82	98.22±0.92	99.10±0.79	98.09±2.58	99.40±0.91	98.86±1.17	**99.55±0.78**
7	97.13±1.63	**99.76±0.16**	99.72±0.43	98.59±2.41	97.93±4.09	99.02±2.47	98.23±3.08	99.12±0.96
8	53.49±28.33	68.54±10.26	69.00±14.31	**77.98±9.03**	73.15±11.71	65.52±12.55	77.95±10.88	77.56±9.58
9	78.63±38.86	92.03±4.81	94.62±2.87	97.48±2.72	98.43±2.56	**99.88±0.24**	98.89±2.05	98.61±1.48
10	23.11±8.71	76.64±7.58	83.52±5.33	76.62±13.06	82.57±6.77	**89.53±8.54**	85.46±11.91	85.62±12.38
11	65.60±18.56	91.64±5.97	95.73±4.96	98.25±0.75	97.34±2.05	98.27±2.38	97.23±2.49	**98.52±1.09**
12	98.07±5.20	97.49±3.57	99.73±0.44	99.24±0.71	**99.77±0.27**	97.60±1.89	98.19±2.38	99.44±0.78
13	**99.48±0.23**	97.37±3.32	97.84±2.21	98.70±1.56	99.05±1.02	98.91±1.11	99.08±1.00	99.32±0.71
14	96.91±1.17	96.99±2.49	98.67±0.58	97.57±2.32	98.83±0.77	**99.28±0.52**	98.35±0.95	98.34±0.74
15	55.94±30.60	68.13±7.92	63.21±14.24	74.74±5.04	77.82±8.59	74.35±10.60	74.43±10.60	**78.32±8.58**
16	73.34±4.68	88.20±8.17	88.02±7.39	88.18±8.42	88.71±6.44	**97.65±1.19**	94.27±5.54	90.51±7.11
AA	76.83±2.09	90.78±0.81	92.09±1.31	92.95±0.87	93.50±1.18	94.18±0.99	93.74±1.03	**94.18±0.87**
OA	70.42±5.13	84.90±1.05	85.63±2.55	88.78±1.80	88.75±2.29	87.86±1.48	89.64±1.53	**90.14±1.17**
Kappa	67.37±5.29	83.25±1.14	84.05±2.80	87.54±1.98	87.52±2.52	86.54±1.61	88.48±1.68	**89.05±1.28**

TABLE V
CLASSIFICATION RESULTS (%) OF VARIOUS METHODS WITH FIVE LABELED SAMPLES PER CLASS ON THE IP DATASET

No.	SVM [11]	HybridSN [37]	Fast3DCNN [38]	DCFSL [36]	Gia-CFSL [48]	DM-MRN [45]	Grid-TF [47]	**Ours**
1	77.56±15.91	77.31±21.27	**97.32±7.27**	92.44±12.83	92.44±13.38	87.07±12.62	96.83±3.28	94.88±9.78
2	27.08±10.68	40.40±9.57	45.37±11.25	38.92±7.94	49.73±9.32	45.57±10.25	48.66±9.36	**53.71±13.20**
3	25.86±7.67	37.31±11.91	51.70±9.52	45.57±5.84	53.38±9.81	46.73±10.87	53.33±9.11	**58.13±5.72**
4	34.27±14.98	40.82±15.85	67.28±14.11	67.50±14.90	**81.51±13.13**	84.13±13.35	78.66±9.90	81.47±11.36
5	38.70±15.15	59.16±11.92	66.21±10.43	67.82±6.48	75.65±7.88	72.86±6.40	74.56±8.44	**76.84±8.99**
6	64.07±15.59	61.97±14.56	87.77±8.85	76.68±8.12	82.87±10.12	**94.27±3.50**	87.38±9.32	87.03±5.64
7	96.52±7.73	88.70±13.78	**100.00±0.00**	98.26±3.98	99.13±2.61	98.26±3.98	99.56±1.30	99.56±1.30
8	91.71±3.79	82.37±12.21	90.74±10.78	88.14±8.79	92.60±5.31	83.91±11.00	92.30±11.93	**94.31±7.37**
9	97.33±5.33	98.67±2.67	**100.00±0.00**	**100.00±0.00**	99.33±2.00	**100.00±0.00**	97.33±4.42	98.67±4.00
10	44.15±21.00	50.76±11.26	55.90±12.45	61.59±10.80	63.06±8.75	**66.70±6.23**	61.42±7.72	66.02±8.45
11	47.98±23.45	53.34±16.50	54.16±16.59	59.19±12.79	60.90±13.45	57.26±14.02	**61.03±11.12**	59.70±9.82
12	26.45±7.59	30.73±10.02	48.89±11.83	35.58±8.37	49.05±11.66	41.24±7.83	49.66±15.74	**56.92±14.13**
13	71.80±18.15	96.80±3.89	98.80±1.76	98.20±2.99	99.25±1.35	**99.40±1.04**	99.00±1.02	98.20±2.30
14	63.44±25.03	74.60±17.89	81.33±10.72	83.06±10.51	81.49±8.52	**93.55±5.59**	84.84±9.28	84.66±10.13
15	50.21±13.67	64.28±11.05	74.33±7.13	56.88±10.73	68.58±12.04	61.36±8.47	78.61±9.13	**83.17±10.73**
16	95.57±5.32	95.68±5.77	99.09±0.99	98.98±2.35	98.98±1.72	**99.65±0.52**	98.98±1.64	99.54±0.75
AA	59.54±2.08	65.81±2.01	76.18±1.37	73.05±1.73	78.00±2.29	77.00±2.25	78.89±1.25	**80.80±0.95**
OA	47.26±4.64	55.14±2.75	63.29±2.86	61.57±2.77	66.64±3.97	66.09±3.22	67.39±2.64	**69.40±1.96**
Kappa	40.76±4.47	49.30±2.76	58.65±2.94	56.59±2.96	62.44±4.23	61.98±3.34	63.27±2.74	**65.51±2.19**

Table IV reports the classification accuracy on the SA dataset under the five-shot condition. Our approach successfully creates a state-of-the-art HSIC accuracy. Similar to the UP dataset, DM-MRN achieves superior performance on AA for the SA dataset, but its performance in OA and Kappa is less satisfactory. Since DCFSL, Gia-CFSL, and Grid-TF also attempt to focus on domain alignment, competitive performance is achieved. Our proposed method clearly outperforms other state-of-the-art methods, improving by about 1.5% on overall accuracy compared to DCFSL and Gia-CFSL. In particular, for "vineyard_untrained" (Class 15) in different states, our approach consistently beats other methods, which implies that the learned class distribution through the cross-domain interaction can facilitate the networks to make more accurate classification of ground objects.

Table V summarizes the classification results on the IP dataset under the five-shot condition. Due to its highly similar spectral information and significant imbalance in sample categories, it becomes challenging to distinguish features for each category accurately. From the performance of CA, one can see that our approach obtains the optimal results on six classes, which benefit from the sufficient cross-class information interaction between the support set and the query set. In terms of AA, OA, and Kappa, our approach consistently produces the optimal HSIC results. In concert, our approach exceeds the second-best Grid-TF method by 2.42%, 2.98%, and 3.54%, respectively. Compared with DCFSL and Gia-CFSL, our approach has exceeded 12.71% and 5.01% on the OA, respectively, further verifying the superiority of our approach in dealing with cross-domain challenges. Note also that our approach exhibits smaller classification variance, demonstrating its superior robustness to perturbations from few-shot sample changes (see, Fig. 1).

Table VI shows the experimental results on the Houston 2018 dataset. Compared to the UP and SA datasets, the HT18 dataset is larger and contains more details and more new categories (e.g., urban infrastructure) to be classified, which brings greater challenges to the classifier. As can be seen from

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TABLE VI

CLASSIFICATION RESULTS (%) OF VARIOUS METHODS WITH FIVE LABELED SAMPLES PER CLASS ON THE HOUSTON 2018 DATASET

No.	SVM [11]	HybridSN [37]	Fast3DCNN [38]	DCFSL [36]	Gia-CFSL [48]	DM-MRN [45]	Grid-TF [47]	**Ours**
1	80.26±12.36	**85.06±8.96**	81.32±10.85	83.55±8.92	82.54±11.59	73.62±2.58	66.67±20.97	72.89±17.04
2	39.65±12.59	44.25±12.54	53.70±14.80	56.72±14.04	59.37±13.60	**64.76±6.21**	59.97±7.00	58.13±13.70
3	84.58±14.99	84.71±8.78	87.98±6.44	95.01±4.27	95.66±5.69	**99.35±1.11**	97.70±2.00	98.39±1.39
4	69.46±12.35	74.27±9.96	73.99±17.91	87.32±6.70	**87.52±4.88**	82.33±16.28	86.61±4.23	86.77±5.22
5	46.59±12.67	39.03±15.70	43.39±16.72	59.49±8.35	64.14±7.51	66.34±9.33	**67.25±5.71**	65.64±8.02
6	35.54±13.47	44.61±18.51	37.20±12.24	75.36±13.64	77.82±13.58	71.05±21.51	78.58±13.58	**84.84±13.06**
7	67.39±17.30	62.72±15.20	64.87±12.04	78.85±17.00	88.70±6.50	99.32±0.56	96.67±6.89	**99.35±1.24**
8	53.36±22.23	54.15±16.99	52.91±18.09	**65.48±11.75**	64.47±13.82	58.76±13.59	60.61±9.09	56.65±15.41
9	19.27±11.42	24.78±5.71	20.33±7.69	22.50±6.98	24.51±7.36	30.77±9.99	32.32±10.55	**32.41±9.48**
10	5.93±5.06	14.88±6.36	14.26±8.76	16.97±5.76	20.02±8.20	23.96±5.70	23.30±8.29	**25.82±5.51**
11	13.53±8.12	12.78±5.08	9.96±2.59	17.19±5.35	26.74±8.31	16.97±5.51	26.03±7.60	**29.53±4.30**
12	28.49±12.05	21.11±6.24	23.83±9.56	32.40±5.93	35.94±10.98	**38.73±9.65**	33.73±8.75	35.08±8.01
13	19.71±11.54	19.00±5.99	17.56±7.79	22.98±8.06	23.70±6.85	26.11±11.58	27.37±10.77	**32.09±9.77**
14	65.79±16.16	51.07±12.74	54.62±13.13	77.56±4.92	75.07±5.03	**85.95±5.62**	74.88±6.24	77.03±6.72
15	88.35±3.32	74.32±12.61	84.36±7.21	**94.95±1.80**	93.98±1.19	92.32±7.96	89.86±8.29	90.43±7.93
16	4.25±2.83	22.80±7.27	19.33±10.00	60.05±10.02	**72.57±7.76**	69.49±13.72	69.80±13.20	71.21±7.46
17	96.74±2.54	84.11±13.82	85.18±14.81	96.45±6.17	94.47±10.40	99.55±1.17	99.43±1.14	**99.72±0.65**
18	26.63±11.50	24.50±8.30	20.45±8.07	57.71±10.28	73.04±11.63	**79.90±12.91**	73.99±9.62	78.74±7.50
19	19.35±14.11	36.77±17.70	31.85±15.93	74.69±9.14	73.49±11.27	78.67±19.68	68.53±14.92	**84.46±10.35**
20	53.91±7.29	52.19±7.95	26.31±13.02	72.41±4.90	75.04±7.06	**84.95±5.38**	79.11±6.24	81.71±5.22
AA	45.94±1.65	46.36±2.65	45.17±2.73	62.38±1.68	65.44±1.62	67.15±2.22	65.62±2.00	**68.04±0.78**
OA	26.90±3.76	30.68±3.62	28.42±3.52	35.83±3.98	38.31±3.74	40.97±3.43	42.27±5.03	**42.72±4.14**
Kappa	21.76±2.33	23.71±3.57	22.23±3.16	30.14±3.59	32.44±3.27	34.55±2.58	35.62±4.21	**36.14±3.50**

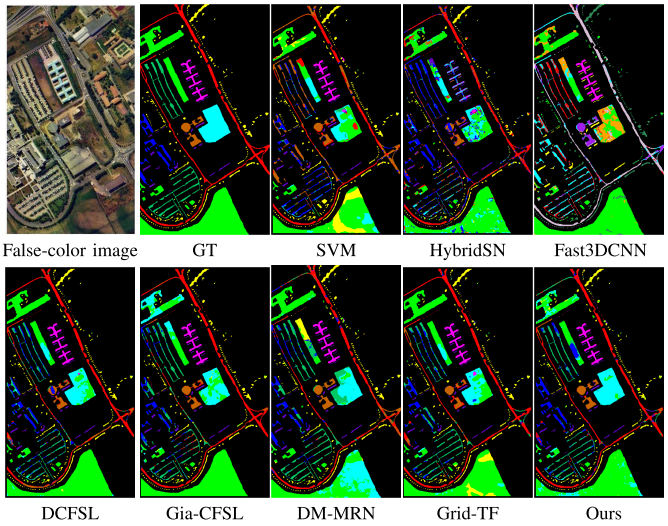


Fig. 5. Visual comparisons against the existing methods on the UP dataset.

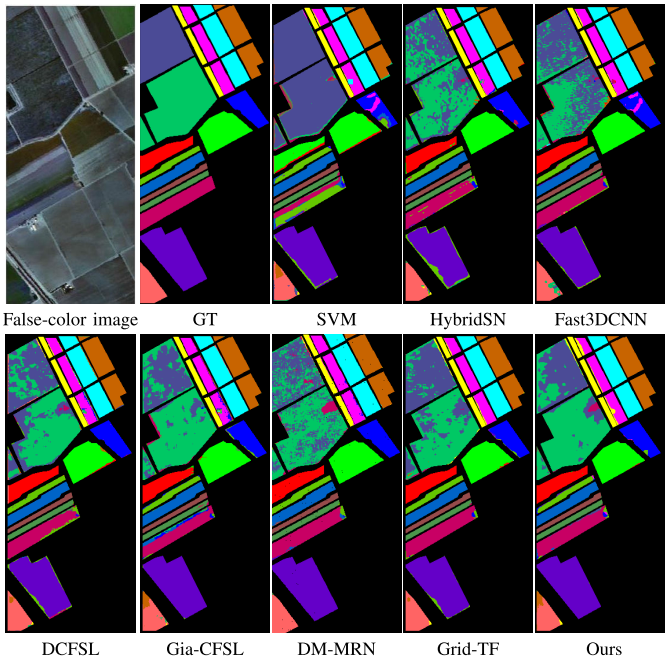


Fig. 6. Visual comparisons against the existing methods on the SA dataset.

the results, our approach has a good detail classification effect and has the highest accuracy in most categories. Furthermore, compared with Grid-TF, the OA, AA, and Kappa are improved by 3.69%, 1.06%, and 1.46%, respectively. These results demonstrate that our approach is conducive to achieving the most superior classification performance, which benefits from cross-domain and cross-class information interactions.

For a more intuitive comparison, we visualize representative classification maps of different methods on the target domain dataset, as depicted in Figs. 5–8. To visually highlight the classification performance, we simultaneously illustrate the corresponding false-color image and ground truth (GT). In Fig. 5, our approach produces more accurate classification results with shaper edges and richer contents. For instance, "Meadows" (Class 2) and "Self-blocking Bricks" (Class 8) have clearer edge structures and are closer to GT. Furthermore, in Fig. 6, "Grapes_untrained" (Class 8) and "Vinyard_untrained" (Class 15) are prone to suffer from misclassification in the comparison methods due to spectral similarity. In contrast, the classification maps generated by our approach have higher fidelity. Moreover, it can be seen from Fig. 7 that our approach achieves satisfactory classification performance and retains more complete details, such as "Woods" (Class 14). Even for the more challenging "Non-residential buildings" (Class 9) in the HT18 dataset, which belongs to the urban infrastructure and is completely different from the categories in the source domain datasets, our approach can still achieve the best classification effect, as exemplified in Fig. 8.

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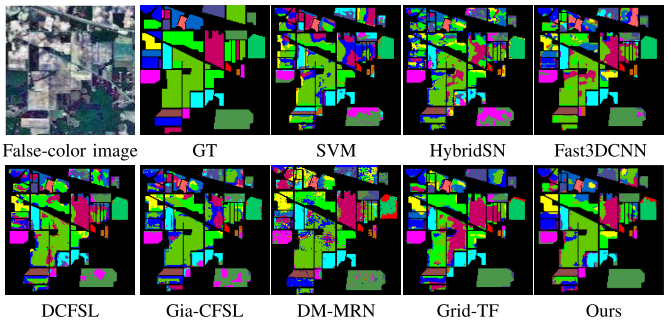


Fig. 7. Visual comparisons against the existing methods on the IP dataset.

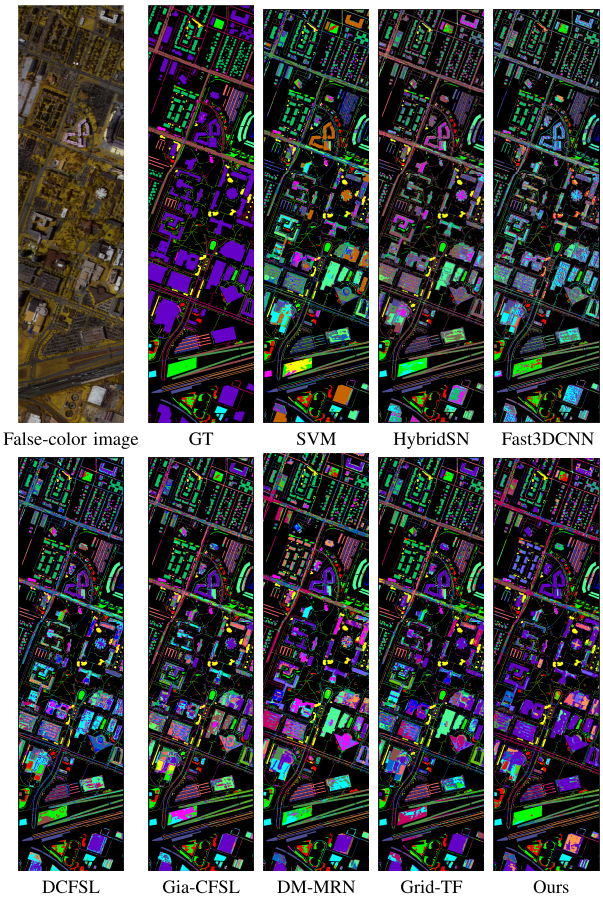


Fig. 8. Visual comparisons against the existing methods on the HT18 dataset.

D. Evaluation of Generalization Capability

In Section IV-C, we have demonstrated the effectiveness of our approach in the case of using Grid-TF [47] as the HSI feature extraction baseline. To further validate the generalization ability of our approach across different HSI FEs, we replace the HSI feature extraction baseline with FEs of HybridSN [37] and Fast3DCNN [38], respectively. It is worth noting that to ensure a fair comparison, we adopt the same experimental settings and evaluation criteria. Fig. 9(a) and (b) presents the experimental results based on HybridSN and Fast3DCNN, respectively. It can be observed that our proposed DA-CAA can consistently improve the classification performance of the existing network models on all three datasets, thanks to our transformation of sample-wise features into class-level features to fulfill robust feature aggregation. Overall, these experiments display that our approach is not limited to specific HSI FEs, showing excellent flexibility and adaptability.

In addition to its ability to adapt to different HSI FEs, our approach also has the potential to be robustly compatible with various few-shot conditions. Fig. 10 illustrates the classification accuracy from one- to five-shot (i.e., $K = \{1, 2, 3, 4, 5\}$) on three benchmark datasets. It can be observed that the classification accuracy increases with the number of labeled novel-class samples. In addition, our approach consistently produces superior classification performance across all few-shot conditions, which is attributed to the fact that our approach can efficiently and adaptively transfer class-level distribution information to the query feature of each novel-class sample, facilitating cross-domain, and cross-class interactions.

TABLE VII

CLASSIFICATION PERFORMANCE (%) OF MODULE FREEZING DURING FINE-TUNING ON THE SA DATASET, INCLUDING THE HSI FE AND THE DAFE MODULE. GRAY DENOTES THE FULL MODEL

Freeze FE	Freeze DAFE	AA	OA	Kappa
✓	✓	90.25±1.14	86.64±1.90	85.16±2.07
✓	✗	90.41±0.85	86.70±1.72	85.23±1.85
✗	✗	93.71±1.28	89.86±1.96	88.74±2.16
✗	✓	**94.18±0.87**	**90.14±1.17**	**89.05±1.28**

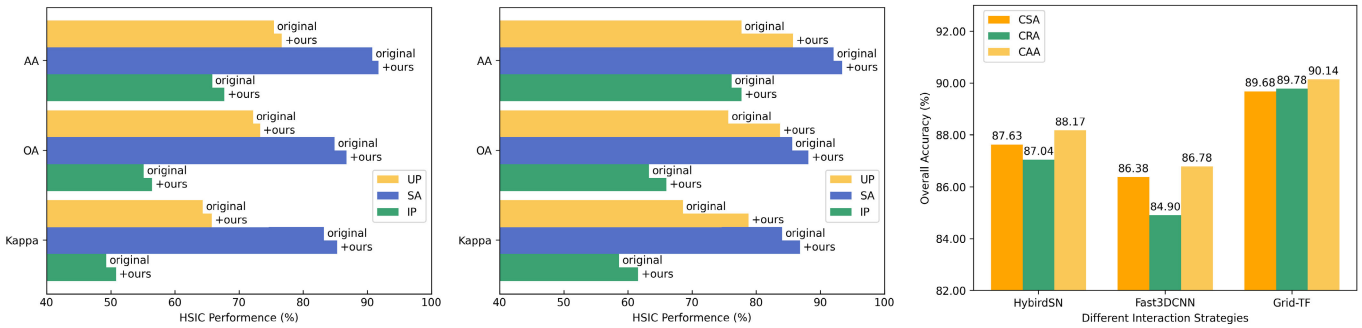
E. Ablation Studies

We present a detailed ablation study of the proposed architecture design in terms of feature aggregation, module freezing, feature dimension, DAFE module, and metric calculation. In the following, we explain each of them in detail.

1) Ablations on Feature Aggregation: We demonstrate the three feature aggregation strategies on the SA dataset in Fig. 9(c), where we apply CSA, CRA, and CAA to three HSIC models which are based on the HSI feature extraction baselines of HybridSN, Fast3DCNN, and Grid-TF, respectively. It is evident that our proposed CAA can effectively enhance HSIC accuracy under few-shot conditions. CSA and CRA solely provide limited complementary information during support–query interactions, thus hindering the advancement of classification performance. In contrast, our CAA models the interaction between distinct classes, integrating the common advantages of CSA and CRA. This enables the adaptive consideration of interclass relationships during classification decisions, thereby helping alleviate class confusion.

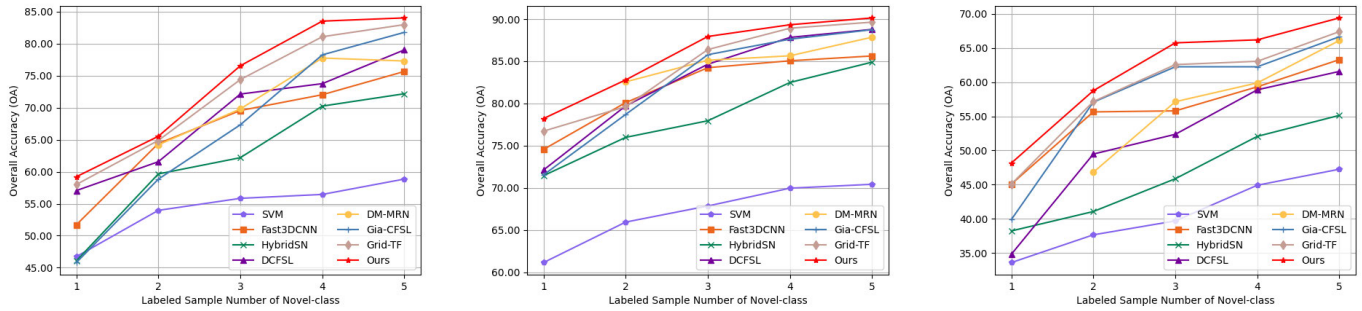
2) Ablations on Training Strategy: After the model is trained on the base-class samples, we fine-tune it on the novel-class samples. At this point, we freeze the parameters of DAFE to ensure that the prior knowledge learned from the base-class samples is maintained. In Table VII, we report quantitative results based on the SA dataset to demonstrate the effects of freezing the HSI FE and DAFE during fine-tuning. It can be observed that freezing the HSI FE significantly degrades the classification performance. In contrast, freezing DAFE is beneficial, indicating that in this scenario, the prior class distribution knowledge learned from abundant base-class samples can be better propagated to novel classes, thereby facilitating the classification performance.

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(a) (b) (c)

Fig. 9. HSIC performance comparison regarding different HSI FEs and aggregation strategies. (a) and (b) HSIC performance on UP, SA, and IP datasets when using HybridSN and Fast3DCNN as HSI FEs, respectively, where "+ours" denotes further integrating our DA-CAA on the original methods. (c) Overall accuracy performance on the SA dataset, with CSA, CRA, and CAA interaction strategies applied to "+ours." Note that here "+ours" represents that the feature extraction baselines of HybridSN, Fast3DCNN, and Grid-TF are adapted as our HSI FE.



(a) (b) (c)

Fig. 10. Comparison of classification performance (%) of various methods from 1- to 5-shot on the target domain datasets. Our approach demonstrates outstanding classification performance across various few-shot conditions. (a) UP dataset. (b) SA dataset. (c) IP dataset.

TABLE VIII

CLASSIFICATION PERFORMANCE (%) WITH DIFFERENT DIMENSIONS OF CLASS-LEVEL FEATURES $\mathcal{Z}$. GRAY DENOTES THE FULL MODEL

Dimension		64	128	256	512
UP	AA	85.35±1.56	**85.87±2.50**	85.28±2.40	85.75±2.35
	OA	83.56±4.39	**84.03±2.67**	82.91±3.27	83.15±3.95
	Kappa	78.88±4.98	**79.24±3.20**	77.93±3.88	78.24±4.57
SA	AA	93.38±1.45	**94.18±0.87**	93.63±1.15	93.91±1.11
	OA	89.19±2.47	**90.14±1.17**	89.22±2.02	89.75±1.78
	Kappa	88.00±2.71	**89.05±1.28**	88.03±2.21	88.62±1.96
IP	AA	**81.01±1.82**	80.80±0.95	80.43±0.79	80.77±0.90
	OA	69.24±2.69	**69.40±1.96**	69.15±2.74	69.25±1.59
	Kappa	65.35±2.96	**65.51±2.19**	65.31±2.89	65.37±1.65

TABLE IX

CLASSIFICATION PERFORMANCE (%) OF DIFFERENT AGGREGATION OBJECTS IN OUR CAA. "W/$\mathcal{Z}$" REPRESENTS THE USE OF CLASS-LEVEL FEATURES $\mathcal{Z}$ FOR FEATURE AGGREGATION AS IN (10), WHILE "W/ $f^{\mathcal{S}}$" DENOTES THE DIRECT USE OF SUPPORT FEATURES $f^{\mathcal{S}}$ FOR FEATURE AGGREGATION, I.E., REMOVING DAFE. GRAY DENOTES THE FULL MODEL

		w/ $f^{\mathcal{S}}$	w/ $\mathcal{Z}$
UP	AA	84.84±2.14	**85.87±2.50**
	OA	82.12±4.52	**84.03±2.67**
	Kappa	77.00±5.10	**79.24±3.20**
SA	AA	94.17±1.29	**94.18±0.87**
	OA	89.86±1.89	**90.14±1.17**
	Kappa	88.75±2.07	**89.05±1.28**
IP	AA	79.37±1.32	**80.80±0.95**
	OA	67.71±2.81	**69.40±1.96**
	Kappa	63.68±3.01	**65.51±2.19**

3) Ablations on Feature Dimension: Regarding the feature dimension L of the class-level features $\mathcal{Z}$ obtained by sampling from the class distribution $\mathcal{N}$ in the support set, we set L to 64, 128, 256, and 512 for our ablations, respectively. Table VIII presents the quantitative experimental results, where our approach obtains the relatively most accurate classification performance when the feature dimension is set to 128. As a result, L is set to 128 in our full model.

4) Ablations on Cross-Domain and Cross-Class Interaction: In our pipeline, we embed class-level features $\mathcal{Z}$ into query features. To verify the effectiveness of DAFE in modeling class distributions, we remove the DAFE module so that the query features $f^{\mathcal{Q}}$ are directly aggregated with the support features $f^{\mathcal{S}}$ in (10). Moreover, we remove $\mathcal{L}_{\text{DAFE}}$, i.e., produce a latent representation by using only the VAE's encoder, which maintains the same network capacity, such that class-level distribution information is not extracted and transferred. We also make the parameters of DAFE not shared between the source and target domains to mitigate cross-domain information interactions. Tables IX and X summarize the ablation results. The consistent improvement in classification performance indicates that the proposed DAFE facilitates the exploration of more general class-level distribution information, thus enabling complementary interactions between query features and support features. In other words, compared with other latent representations, it is more beneficial to transfer the

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TABLE X

ABLATION RESULT (%) OF THE DAFE MODULE ON THE SA DATASET. "W/O $\mathcal{L}_{\text{DAFE}}$" INDICATES THAT ONLY THE ENCODER OF THE VAE IS RETAINED TO GENERATE A LATENT REPRESENTATION THAT IS NOT CLASS-LEVEL DISTRIBUTION-AWARE; "W/O SHARE" DENOTES THAT THE DAFE USED IN THE SOURCE AND TARGET DOMAINS DOES NOT SHARE PARAMETERS, THEREBY WEAKENING CROSS-DOMAIN INTERACTION

	AA	OA	Kappa
w/o $\mathcal{L}_{\text{DAFE}}$	93.65±1.25	89.84±2.15	88.72±2.36
w/o share	93.87±0.98	89.77±1.97	88.64±2.18
Full model	**94.18±0.87**	**90.14±1.17**	**89.05±1.28**

TABLE XI

CLASSIFICATION COMPARISON (%) ON THE SA DATASET WITH VARYING DISTANCE METRICS, INCLUDING NN, ED, AND MD

	AA	OA	Kappa
w/ NN	92.92±1.23	88.18±1.78	86.89±1.95
w/ ED	93.07±1.19	88.36±1.81	87.09±1.19
w/ MD	**94.18±0.87**	**90.14±1.17**	**89.05±1.28**

TABLE XII

CLASSIFICATION RESULT (%) ON THE SA DATASET WITH VARYING λ_{cls}

λ_{cls}	1	3	5	7	10
AA	93.98±1.03	94.01±0.75	**94.18±0.87**	94.05±0.61	94.02±1.04
OA	89.72±1.92	89.94±1.36	**90.14±1.17**	90.07±1.36	90.01±1.69
Kappa	88.57±2.14	88.83±1.51	**89.05±1.28**	89.01±1.50	88.99±1.88

class-level distribution information extracted by our DAFE across classes between the support set and the query set, and the use of parameter-sharing DAFE also aids in cross-domain information interaction. Consequently, our DAFE is capable of transforming sample-wise features into class-level features, enabling cross-domain and cross-class enhancement for more robust and effective feature aggregation.

5) Ablations on Metric Calculation: The meta-learning methods based on metric learning perform classification by measuring the similarity between sample features. In our pipeline, we chose the MD for metric calculation [see (13)]. To further validate its rationality, we show the classification efficacy using nearest neighbor (NN), and ED in Table XI. With all other settings unchanged, leveraging the MD as the metric calculation significantly achieves the most outstanding classification accuracy. This is because, in our proposed DA-CAA method, the enhanced query features embed distribution information through cross-domain and cross-class information interactions, naturally aligning with the MD which determines decision boundaries by estimating the correlation between classes.

6) Ablations on Loss Function: In our total loss function of (9), we set λ_{cls} to 5. Here, we provide ablation of λ_{cls} on the SA dataset. From the results in Table XII, we can observe that the best classification accuracy can be achieved when λ_{cls} is set to 5. Furthermore, we empirically set λ_{KL} to 2.5×10^{-4} in (8). Note that λ_{KL} is set relatively small similar to [52], [59]. This is because $\mathcal{L}_{\text{rec}}$ is devoted to ensuring that the VAE's encoder can extract more meaningful information for subsequent feature interaction, while $\mathcal{L}_{\text{KL}}$ is used to make z follow a Gaussian distribution. When λ_{KL} is set to 1, we obtain AA, OA, and Kappa of 92.41 ± 0.95, 88.20 ± 1.57, and 86.91 ± 1.72 on the SA dataset, which is a decrease of 1.88%, 2.15%, and 2.40%, respectively, demonstrating that our parameter setting is reasonable.

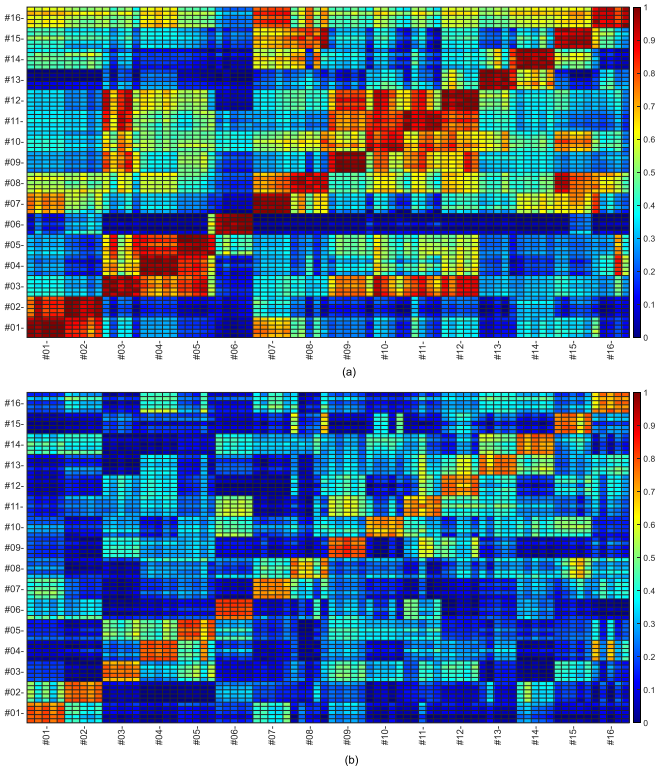


Fig. 11. Visualization of the heatmap of metric calculation on the SA dataset. (a) Based on Grid-TF [47]. (b) Based on our method. Each class contains five labeled support set samples (vertical axis) and five unlabeled query set samples (horizontal axis). Warmer colors indicate higher similarity.

F. Visualization of Metric Calculation and t-SNE

We have displayed the superiority of our approach in Section IV-C. To further prove our approach's ability to mitigate class confusion, we visualize the metric calculation in Fig. 11, where the normalized MDs [see (13)] between 16 classes (five samples per class) are shown. Unlike the vanilla Grid-TF [47], which appears disorderly and interconfusing, our approach concentrates the similarity metric values on the diagonal, indicating clearer classification boundaries between classes. Moreover, we present the t-SNE [63] visualization of features for each class in Fig. 12, where the features generated by our approach are more separable, such as the orange part, leading to more promising classification results. This is because our approach integrates the DAFE and CAA modules to facilitate interaction between source and target domains as well as support and query sets. Therefore, the enhanced query features contribute to increasing the discrimination between features during metric calculation, i.e., greatly reducing class confusion.

G. Analysis of Network Size and Inference Efficiency

We conduct a statistical analysis of the network parameters and inference time for the traditional SVM method and deep learning-based DCFSL, Gia-CFSL, DM-MRN, and Grid-TF

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TABLE XIII

COMPARISON OF NETWORK PARAMETERS (MILLION) AND INFERENCE TIME (SECOND) OF DIFFERENT METHODS

Method		SVM	DCFSL	Gia-CFSL	DM-MRN	Grid-TF	Ours
#Param.		–	4.26	0.31	0.18	0.95	1.01
#Runtime	UP	0.96	11.04	12.1	49.93	5.96	20.87
	SA	1.21	13.78	15.54	114.27	6.36	26.88
	IP	0.31	2.76	3.52	36.18	1.43	5.08

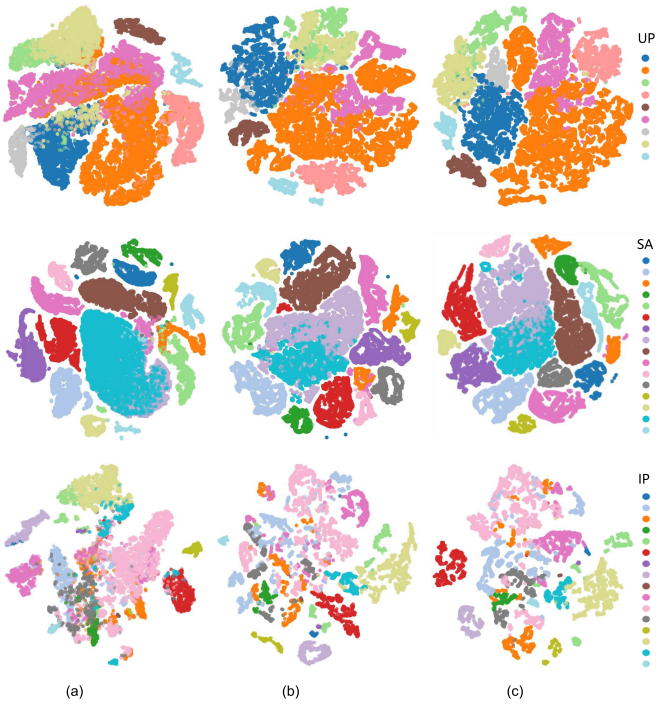


Fig. 12. t-SNE visualization of features for each class. The results on the UP, SA, and IP datasets are from top to bottom, respectively. The legend on the right uses different colors to represent different categories in sequence. (a) DM-MRN. (b) Grid-TF. (c) Ours.

methods, as shown in Table XIII. Note that DM-MRN has the longest running time. Due to the use of transformer [64], [65], [66] to establish long-range spectral dependencies, Grid-TF has a relatively larger parameter size. Notably, our pipeline utilizes Grid-TF as the HSI FE, which only adds a very small number of model parameters (0.06 MB). Therefore, our DA-CAA is lightweight and effective, making it flexible for improving classification performance with varying HSI feature extraction baselines, as demonstrated in Section IV-D. Although our proposed strategy has been proven effective in boosting classification performance, it introduces additional inference time due to the emphasis on aggregating each sample-specific query feature with class-level features (involving distribution information of classes) of all support set samples. In contrast, CSA and CRA enable each query feature to interact with the class-level features of a specific/arbitrary class instead of all classes, resulting in less inference time, i.e., 14.96, 19.05, and 3.70 s on the UP, SA, and IP datasets, respectively. Designing more efficient feature interaction schemes to improve inference speed will be an area for future improvement, such as making each query feature interact with the support features of the top-k classes with the highest similarity (typically, $k = 1$ for CSA and CRA and $k = C$ for our CAA).

V. CONCLUSION

In this article, we propose a flexible and effective few-shot HSIC method by transferring the class-level distribution information learned from the base classes to the novel classes. To begin with, we develop a tractable DAFE, which employs the widely used VAE pretrained on abundant-labeled base-class samples to encode the support set samples as class distributions. Subsequently, we sample class-level features from the learned distribution. Finally, we design a CAA module to efficiently aggregate them with sample-specific query features, which upgrades the reliability of model predictions through sufficient cross-domain and cross-class feature interactions in the meta-task. Experimental results show that our method achieves superior few-shot HSIC performance compared to state-of-the-art methods. Moreover, it can effectively enhance the classification capability of the existing models by harnessing its flexibility across various HSI FEs. In the future, it is worthwhile to utilize generative models (e.g., diffusion [67], [68], [69]) to obtain more representative class-level features, or borrow vision foundation models [70], [71], [72] to embed more comprehensive prior knowledge to further improve the model performance.

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